Exercice 1

1

$$A = \frac{1 + \dfrac{1}{1 - \dfrac{1}{x}}}{1 - \dfrac{1}{1 - \dfrac{1}{1 - \dfrac{1}{x}}}} = \frac{1 + \dfrac{x}{x-1}}{1 - \dfrac{1}{1 - \dfrac{x}{x-1}}}$$

$$= \frac{\dfrac{x-1}{x-1} + \dfrac{x}{x-1}}{1 - \dfrac{1}{-\dfrac{1}{x-1}}} = \frac{\dfrac{2x-1}{x-1}}{1 + (x-1)}$$

$$= \frac{\dfrac{2x-1}{x-1}}{x} = \frac{2x-1}{x(x-1)}.$$

$$\boxed{A = \frac{2x-1}{x(x-1)}.}$$

2

$$E = \frac{b + \dfrac{b^2}{b-a}}{b - \dfrac{b^2}{b-a}} = \frac{\dfrac{b(b-a) + b^2}{b-a}}{\dfrac{b(b-a) - b^2}{b-a}} = \frac{b^2 - ab + b^2}{b^2 - ab - b^2}$$

$$= \frac{2b^2 - ab}{-ab} = \frac{b(2b-a)}{-ab} = -\frac{2b-a}{a} = \frac{a-2b}{a}.$$

$$\boxed{E = \frac{a-2b}{a}.}$$

Exercice 2

① $\sqrt{\frac{75}{27}} = \sqrt{\frac{25\times 3}{9\times 3}} = \sqrt{\frac{25}{9}} = \boxed{\frac{5}{3}}$

② $\sqrt{\frac{28}{14}} = \sqrt{\frac{2\times 14}{14}} = \boxed{\sqrt{2}}$

③ $\left(\sqrt{3}+\sqrt{5}\right)^2 + \left(\sqrt{15}-1\right)^2$

$= \left(3+5+2\sqrt{15}\right) + \left(15+1-2\sqrt{15}\right)$

$= 8+16$

$= \boxed{24}$

④ $\left(\sqrt{3}+\sqrt{6}\right)\left(1-\sqrt{2}\right)$

$= \sqrt{3} - \sqrt{3\times 2} + \sqrt{6} - \sqrt{6\times 2}$

$= \sqrt{3} - \sqrt{6} + \sqrt{6} - \sqrt{12}$

$= \sqrt{3} - \sqrt{4\times 3} = \sqrt{3} - 2\sqrt{3}$

$= \boxed{-\sqrt{3}}$

⑤ $\left(\sqrt{7}+2\sqrt{3}\right)\left(1-\sqrt{5}\right)$

$= \sqrt{7} - \sqrt{35} + 2\sqrt{3} - 2\sqrt{15}$

$\boxed{\sqrt{7}+2\sqrt{3}-\sqrt{35}-2\sqrt{15}}$

⑥ $3\sqrt{20} + 4\sqrt{45} - 2\sqrt{80} - \sqrt{180}$

$= 3\times 2\sqrt{5} + 4\times 3\sqrt{5} - 2\times 4\sqrt{5} - 6\sqrt{5}$

$= 6\sqrt{5} + 12\sqrt{5} - 8\sqrt{5} - 6\sqrt{5}$

$= \boxed{4\sqrt{5}}$

⑦ $5\sqrt{12} + 8\sqrt{27} + \sqrt{75} - 2\sqrt{48} - \sqrt{147}$

$= 5\times 2\sqrt{3} + 8\times 3\sqrt{3} + 5\sqrt{3} - 2\times 4\sqrt{3} - 7\sqrt{3}$

$= 10\sqrt{3} + 24\sqrt{3} + 5\sqrt{3} - 8\sqrt{3} - 7\sqrt{3}$

$= \boxed{24\sqrt{3}}$

⑧ $\left(\sqrt{28}+\sqrt{7}-\sqrt{32}\right)\left(\sqrt{63}+2\sqrt{8}\right)$

$= \left(2\sqrt{7}+\sqrt{7}-4\sqrt{2}\right)\left(3\sqrt{7}+2\times 2\sqrt{2}\right)$

$= \left(3\sqrt{7}-4\sqrt{2}\right)\left(3\sqrt{7}+4\sqrt{2}\right)$

$= \left(3\sqrt{7}\right)^2 - \left(4\sqrt{2}\right)^2$

$= 9\times 7 - 16\times 2 = 63 - 32$

$= \boxed{31}$